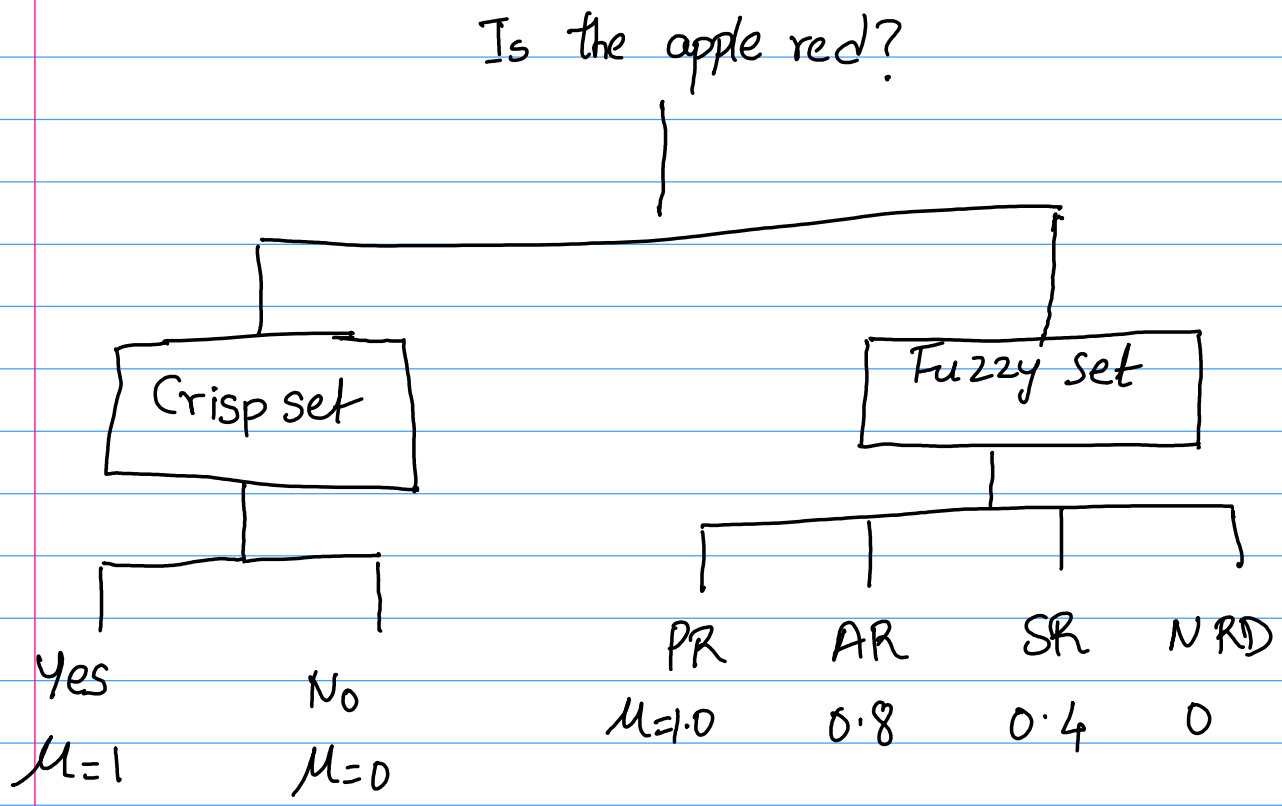
Fuzzy Logic

Crisp sets

Fuzzy sets

X Y

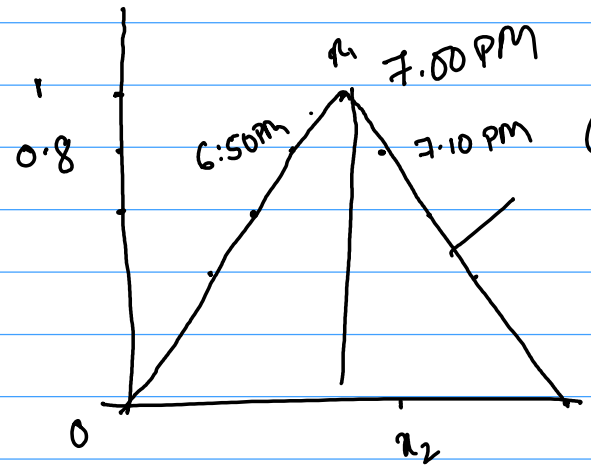


PR = Perfectly red $\mu = 1.0$

AR = Almost red $\mu = 0.8$

SR = Slightly red $\mu = 0.4$

NRD = Not red $\mu = 0.0$

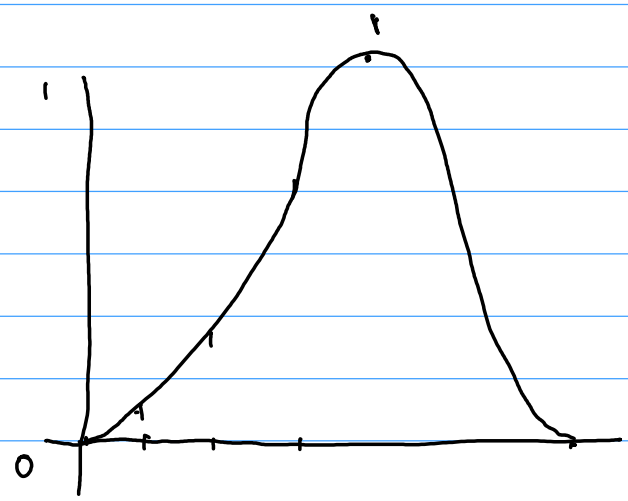


On time

$(x_1, 1)$ ✓

$(x_2, 0.8)$ ✓

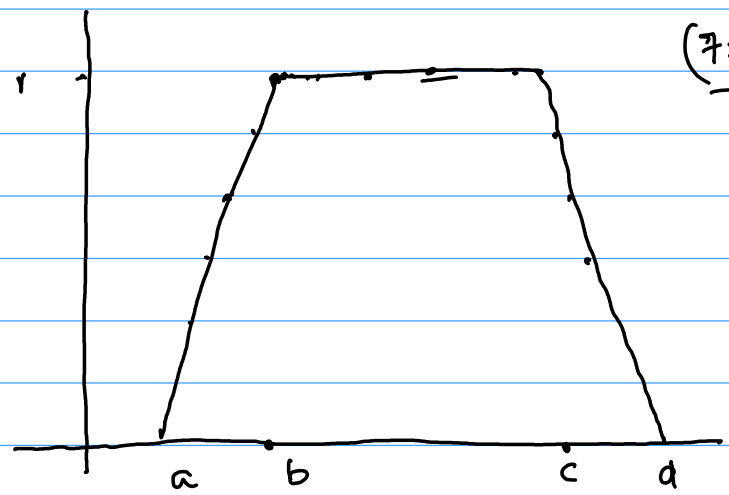
✓



X

Y

(7:00 to 7:20)



Crisp set

Universal set or

Universe of discourse.

5 $\qquad\qquad\qquad\qquad$ X

$A = \{a_1, a_2, a_3, a_4, a_5 \cdots\}$

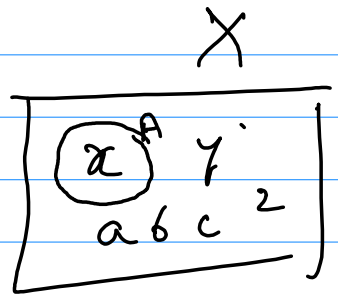
$A = \{x \mid \underline{P(x)}\}$

Notations in set theory

1. ϕ null set or empty set.

2. $\underline{x} \in \underline{A}$

3 $x \notin A$



4 $A \subseteq B$ A is subset of B

5. $A \supseteq B$ A is superset of B

6. $A = B$

7. $A \neq B$

8. $A \subset B$ A is proper subset of B

9. $A \supset B$ A is proper superset of B.

10. $|A|$ Cardinality of A.

$A = \{x_1, x_2, x_3\}$ $|A| = 3$

$A = \{a_1\}$ $|A| = 1$

11 $p(A)$ power set of A

$A = \{a_1, a_2, a_3\}$ $2^3 = 8$

$A = \{a_1, a_2, a_3\}$

$P(A) = 8$ power set of A

2^3 $\{a_1\}$ $\{a_2\}$ $\{a_3\}$ $\{a_1, a_2\}$ $\{a_2, a_3\}$

$\{a_3, a_1\}$ $\{\phi\}$ $\{a_1 a_2 a_3\}$

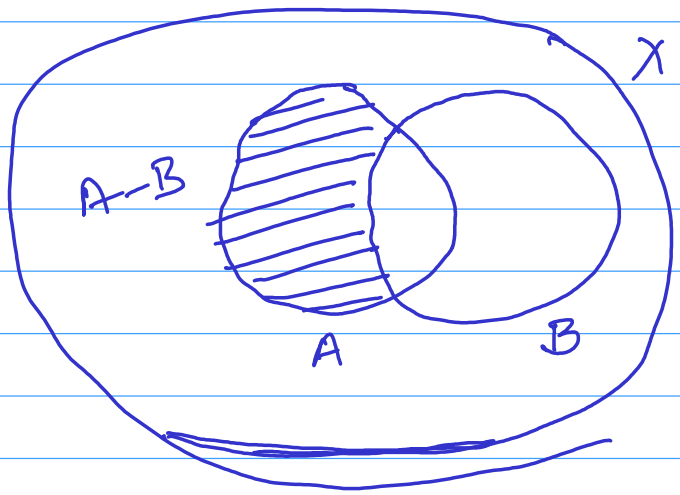
$A = \{a_1 a_2\}$ $2^2 = 4$

$\{a_1\}$ $\{a_2\}$ $\{a_1, a_2\}$ $\{\phi\}$.

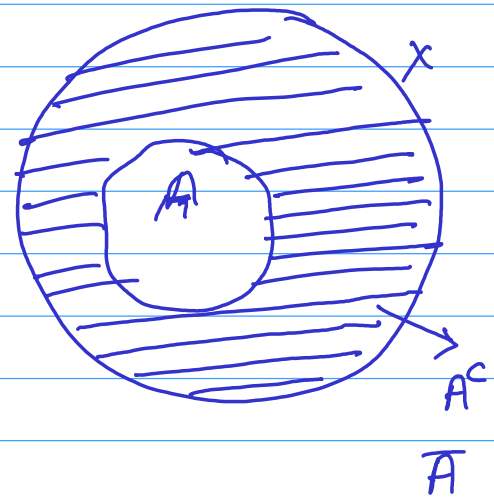
Crisp set operations.

Difference.

$A - B = \{x \mid x \in A \text{ and } x \notin B\}$



$\bar{A} = A^c = X - A$



Absolute component

$A = \{a, c, d, f, r, g, h\}$ and $B = \{c, d, r, m, n\}$

$A - B = \{a, f, g, h\}$

Intersection.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

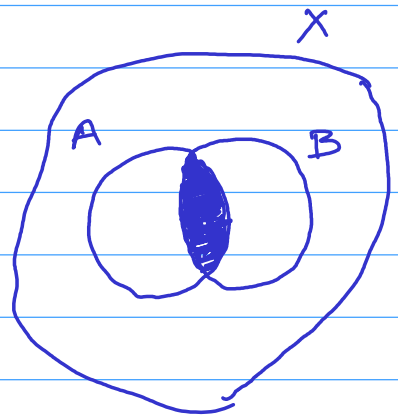
A B as disjoint

$$A \cap B = \emptyset$$

Ex: $A = \{a, c, d, f, r, g, h\}$ and

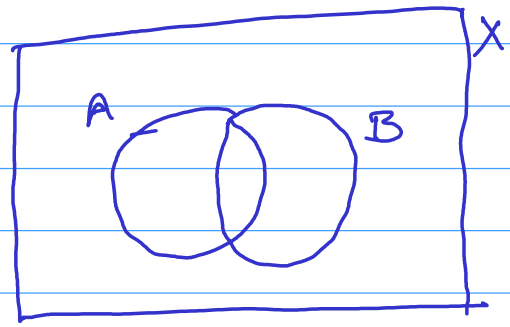
$B = \{c, d, r, m, n\}$

$A \cap B = \{c, d, r\}$



Union:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$



$A = \{a, c, d, f, r, h\}$

$B = \{c, d, r, m, n\}$

$A \cup B = \{a, c, d, f, r, g, h, m, n\}$

Properties of crisp set:

① Law of involution $\bar{\bar{A}} = A$

② Law of commutativity

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

③ Law of associativity

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

④ Law of idempotence

$$A \cup A = A$$
$$A \cap A = A$$

⑤ Law of absorption.

$$A \cup (A \cap B) = A$$
$$A \cap (A \cup B) = A$$

→ Laws of identity

$$A \cup X = X$$

$$A \cap X = A$$

$$A \cup \emptyset = A$$

$$A \cap \emptyset = A$$

→ De Morgan's law

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B}$$

$$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

→ Law of contradiction:

$$A \cap \bar{A} = \emptyset$$

→ Law of excluded middle

$$A \cup \bar{A} = X$$

Representation of fuzzy set

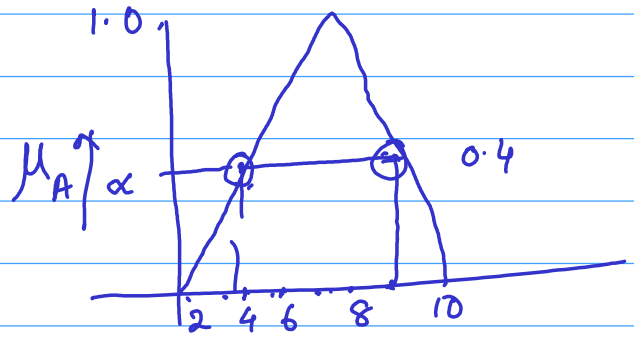
$$A(x) = \{ (x, \mu_A(x)), x \in X \}$$

Discrete fuzzy set.

$$A(x) = \sum_{i=1}^{n} \mu_A(x_i) / x_i$$

Few definitions of fuzzy set:

1. α- Cut of a fuzzy Set



$$\alpha_{\mu_A}(x) = \{x \mid \mu_A(x) \geq \alpha\}$$

Strong $\alpha^+_{\mu_A}(x) = \{x \mid \mu_A(x) > \alpha\}$